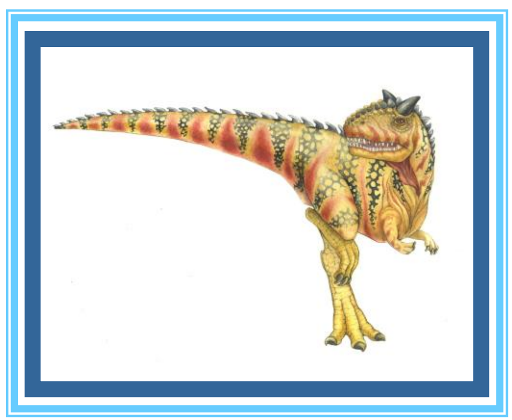
Chapter 4: Threads & Concurrency



Outline

- Overview
- Multicore Programming
- Multithreading Models
- Thread Libraries
- Implicit Threading
- Threading Issues
- Operating System Examples

Objectives

- Identify the basic components of a thread, and contrast threads and processes
- Describe the benefits and challenges of designing multithreaded applications
- Illustrate different approaches to implicit threading including thread pools, fork-join, and Grand Central Dispatch
- Describe how the Windows and Linux operating systems represent threads
- Designing multithreaded applications using the Pthreads, Java, and Windows threading APIs

Motivation

- Most modern applications are multithreaded
- Threads run within application
- Multiple tasks with the application can be implemented by separate threads
  - Update display
  - Fetch data
  - Spell checking
  - Answer a network request
- Process creation is heavy-weight while thread creation is light-weight
- Can simplify code, increase efficiency
- Kernels are generally multithreaded

Aplicatii concurente

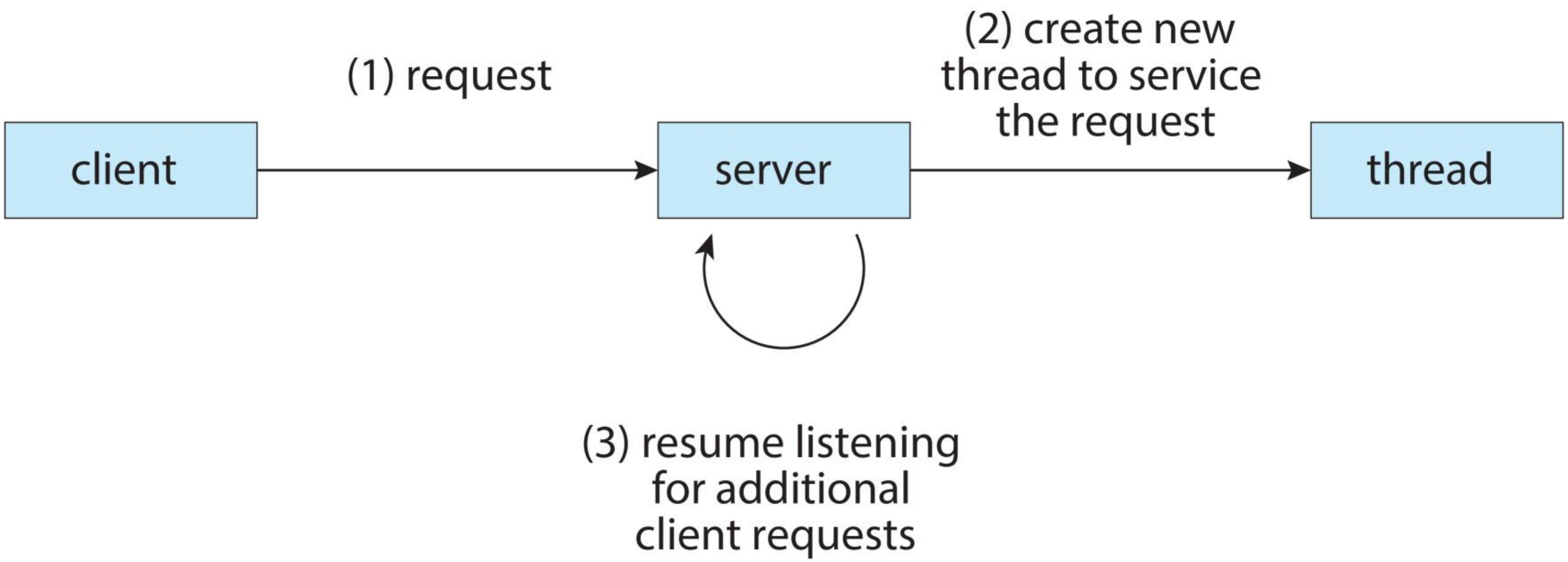
- proces = un singur punct de executie in aplicatie (o singura instanta in rulare a aplicatiei)
- unele aplicatii pot profita de existenta mai multor puncte de executie simultana in cadrul aplicatiei (mai ales pe multiprocesoare)
- ex. 1: procesor cu 4 core-uri + aplicatie de filtrare de imagini care imparte imaginea in 4 cadrane, fiecare core filtrand un cadran
  - avantaj: descompunerea activitarilor mari in activitati mai simple care ruleaza simultan => reducerea timpului de rulare
- ex. 2: un singur procesor + program de gestiune a ferestrelor in GUI
  - verifica si proceseaza input-ul de la tastatura, mouse si retea pt. a produce output pe ecran
  - activitati concurente: verificare/procesare input tastatura, mouse si retea, afisare bitmap-uri pe ecran
    - activitati codate in module separate, cu date private, comunica intre ele prin date partajate

Aplicatii concurente (cont.)

- ex. 3: server de retea
  - activitatea principala: asteapta cereri de la client, le proceseaza si trimite inapoi raspunsurile
  - cu un singur punct de executie procesarea unei cereri particulare ia mult timp (eg, in asteptarea datelor de pe disc)

    => intarzieri mari pt tratarea altor cereri (situatie similara cu cea care a condus la nevoia de multitasking)
  - solutie: fiecare cerere client e procesata in alt punct de executie al aplicatiei server

Multithreaded Server Architecture



Procese si date partajate

- crearea de puncte multiple de executie in program cu procese care partajeaza date (eg, prin memorie partajata, cu acces protejat cu semafoare/locks)
  - procesele au date private (datorita protectiei MMU a spatiului de adresa)
  - comunica intre ele prin IPC (memorie partajata)
  - daca exista capacitare de multiprocesare, procesele pot rula simultan pe mai multe procesoare
- puncte nevralgice
  - crearea proceselor
  - context-switch-ul
  - IPC-ul

Crearea proceselor

- contextul unui proces e stufos (stocat in PCB in kernel)
  - starea completa a CPU
  - registre de gestiunea memoriei
  - tabele de pagini de memorie
  - descriptori de fisiere
  - actiuni asociate semnalelor, etc

    => cost semnificativ de creare a unui proces
- daca acest cost e prea mare => procesoare folosite ineficient
  - ex aplicatie de filtrare imagini
    - pt timp de creare proces comparabil cu timpul de filtrare al unui cadran, filtrarea intregii imagini cu 4 procese poate dura mai mult decat filtrarea cu un singur proces (*slowdown*)

Context-switch

- salvarea contextul unui proces + incarcarea contextului unui nou proces => un cost semnificativ avand in vedere bogatia de informatii din PCB
- ex: secventa de evenimente pt. producator-consumator cu zero buffering (*rendezvous*)
  - producatorul produce un element si se blocheaza
  - sistemul face context-switch si aduce consumatorul pe procesor
  - consumatorul consuma elementul si se blocheaza
  - sistemul face context-switch si aduce producatorul inapoi pe procesor pt a produce un nou element
- timp de context-switch mare => viteza de transfer a datelor intre producator si consumator serios afectata
  - fiecare transfer implica doua context-switch-uri
  - oricum, imposibil sa se atinga viteza teoretica maxima de transfer

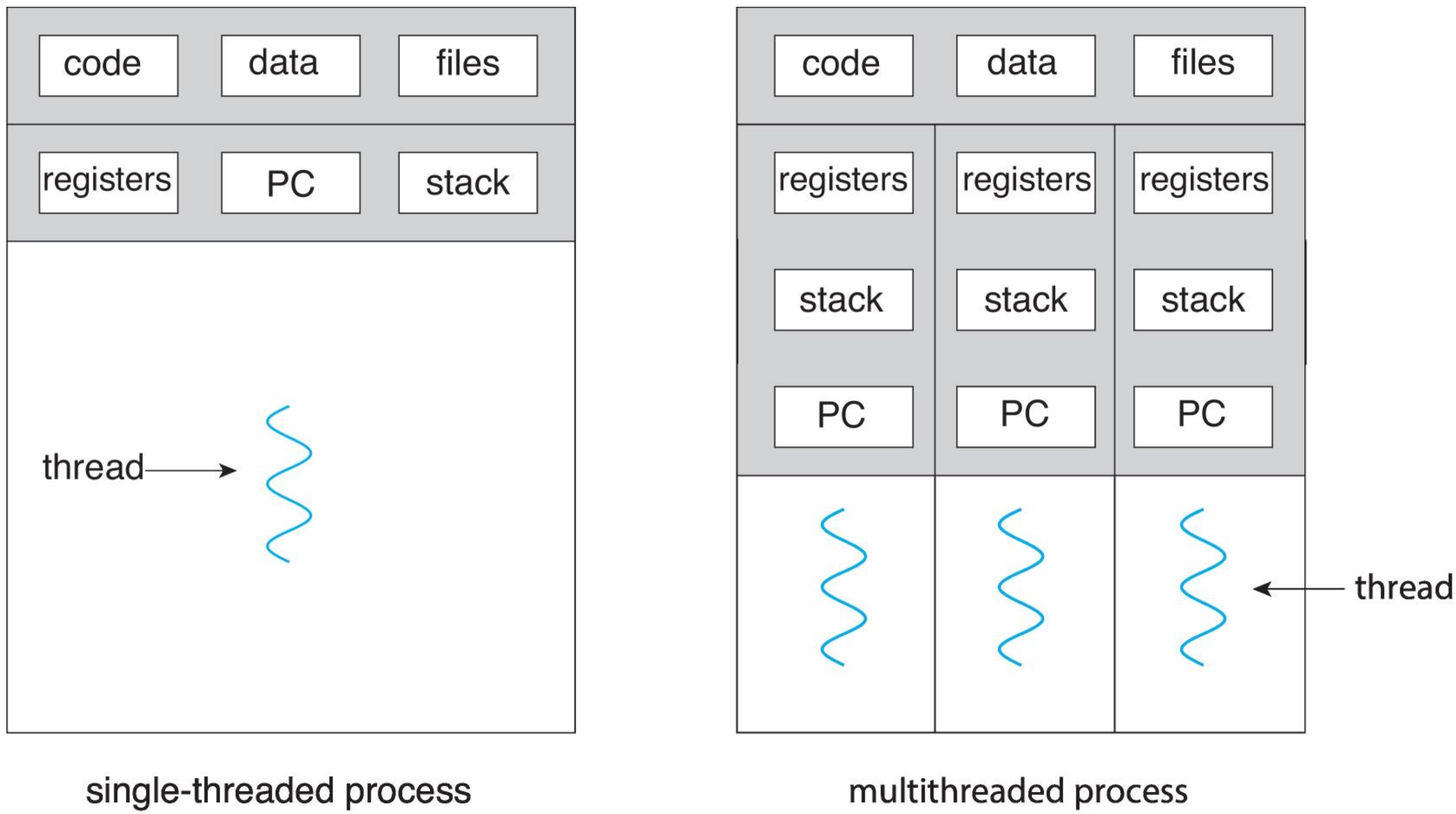
Fire de executie (threads of execution)

- modalitate de a reduce costuri pentru
  - crearea punctelor de executie in aplicatie
  - schimbarii contextelor de executie intre ele
  - partajarea datelor (IPC)
- idee: puncte de executie multiple din aplicatie partajeaza o parte din contextul programului (registrele de gestiune a memoriei, tabelele de pagini, descriptorii de fisiere deschise, samd)
- DAR, fiecare punct de executie are:
  - o copie individuala a unui subset al contextului de executie a aplicatiei (de ex. contextul CPU)
  - structuri de date necesare punctului de executie (de ex. stiva)

=> o noua abstractie de nivel inalt, *firul de executie (thread)* = un punct de executie cu context redus in cadrul programului

- referit uneori din cauza acestei viziuni ca fiind un *proces usor* (lightweight process, LWP)

Single and Multithreaded Processes



Benefits

- **Responsiveness –** may allow continued execution if part of process is blocked, especially important for user interfaces
- **Resource Sharing –** threads share resources of process, easier than shared memory or message passing
- **Economy –** cheaper than process creation, thread switching lower overhead than context switching
- **Scalability –** process can take advantage of multicore architectures

Caracteristici threaduri

- ruleaza secvential, au program counter si stiva proprii
- multiplexeaza accesul la CPU ca si procesele (pe multiprocesoare ruleaza in paralel)
- pot crea alte threaduri
- pot executa apeluri de sistem
  - thread blocat in apel sistem => alt thread primeste procesorul
- analogie posibila
  - threadul este fata de proces ceea ce procesul e in raport cu procesorul
  - procesul actioneaza ca un procesor virtual pe care ruleaza thread-ul

Diferente fata de procese

- threadurile aceluiasi proces partajeaza spatiul de adresa al procesului (de ex, partajeaza variabilele globale)

  => un thread poate distruge usor alt thread (nu exista protectia MMU ca in cazul proceselor diferite)

- lipsa protectiei intre threaduri
  - inevitabila prin design
  - nu e necesara (threadurile sunt parte a aceluiasi program al unui anumit utilizator)
  - nici nu e de dorit (impunerea unor domenii de protectie conduce la probleme similare proceselor)
- alte resurse partajate: acelasi set de fisiere deschise, timere, semnale, etc

Alte caracteristici ale threadurilor

- stari (la fel ca la procese): *in rulare*, *gata de rulare*, *blocat*, *terminat*
- modele de utilizare
  - cooperativ, lucru in echipa (ex filtrare de imagini)
  - master-slave/worker (ex server)
  - pipeline (ex producator-consumator)
- avantaj principal: datele partajate sunt datele globale din proces
  - nu e nevoie de mecanisme IPC de tipul memoriei partajate ca in cazul proceselor
    - ex: buffer global pt. producator-consumator
  - argument puternic pt sistemele multiprocesor unde threadurile pot rula pe CPU-uri diferite

=> partajarea implicita a datelor prin spatiul comun de adresa

Design-ul pachetelor de threaduri

- pachet de threaduri = colectie de primitive (apeluri de biblioteca) pt lucrul cu threaduri
- (1) gestiunea threadurilor
  - *creare thread*:
    - argumente:
      - functia care reprezinta punctul de executie initial
      - stiva privata
      - prioritate de planificare pe procesor
    - intoarce un TID (Thread ID)
  - *terminare thread*: explicit prin apel *thread exit* sau semnal de tip *kill* de la alt thread/proces
  - primitive de sincronizarea accesului la date partajate
    - mutex-uri + variabile de conditie
    - ex: *acquire/release* resurse folosind mutex+variabile conditie

Design-ul pachetelor de threaduri (cont.)

- (2) planificare (aceeasi algoritmi ca si la procese, vom discuta la planificarea proceselor/threadurilor)
- (3) probleme de reentranta
  - scenariu: doua threaduri T1 si T2 executa concurent apeluri sistem, T1 reuseste, T2 esueaza
  - daca T1 nu evalueaza *errno* inainte ca T2 sa execute apelul sistem care esueaza, T1 va crede eronat ca apelul sau sistem a esuat
  - problema principiala: *errno* e variabila globala
  - solutii posibile:

  (a) protejarea *errno* cu mutex-uri

  (b) copie privata a lui *errno* stocata in variabila globala thread-ului apelant, dar privata (invizibila) celorlalte thread-uri

  => TLS (Thread-Local Storage)

Implementarea threadurilor kernel

- pachetele de threaduri se pot implementa in kernel sau in spatiul utilizator
- threadurile kernel separa campurile din PCB care ajuta la crearea unui punct de executie si le stocheaza intr-un TCB

  => un proces cu un singur punct de executie e reprezentat **in kernel** de un PCB si un TCB
- operatia de creare a unui thread (*thread_create*) este un apel sistem
  - aloca un TCB
  - aloca stive kernel si user
  - le leaga la PCB-ul procesului in care s-a facut apelul sistem

Exemplu cu doua threaduri kernel

- un thread suspendat in kernel
- altul ruland in spatiul utilizator
- pt ca sunt implementate in kernel si seamana cu procesele, threadurile kernel se mai cheama si *procese usoare* (*lightweight processes, LWP*)

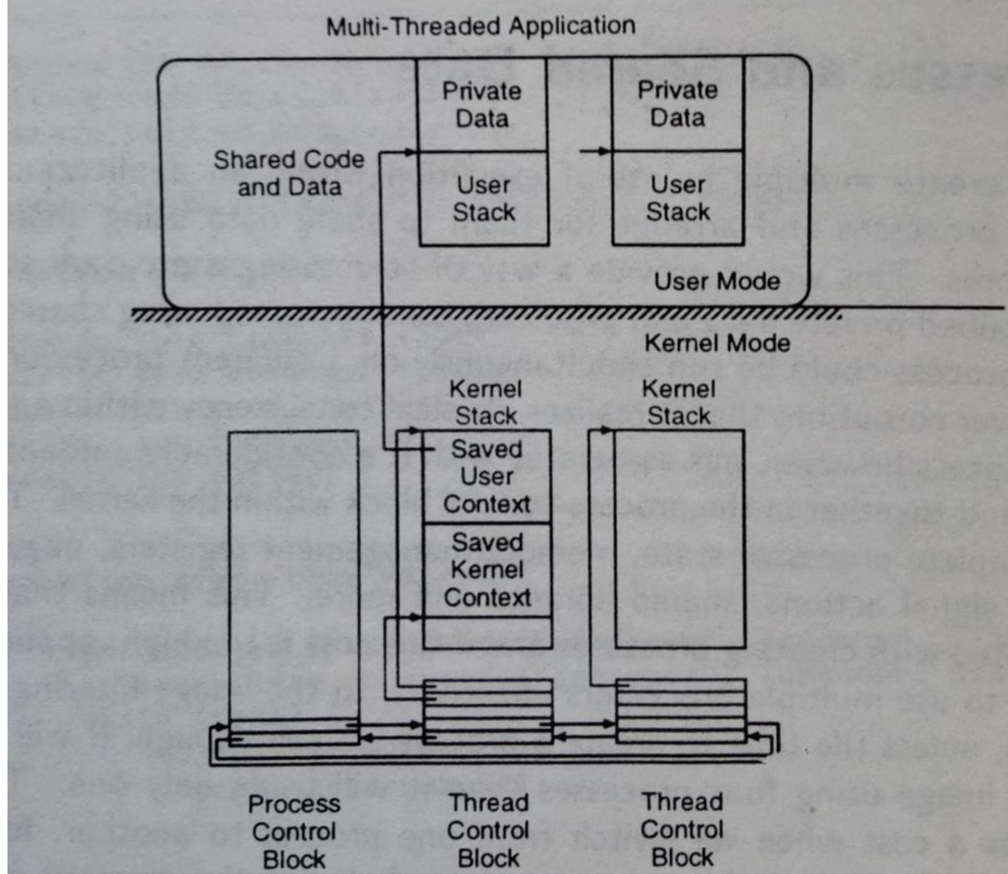


Figure 3: Example of kernel thread implementation.

Costuri threaduri kernel

- cost creare thread kernel << cost creare proces
  - se aloca si initializeaza doar TCB-ul si stivele
  - restul contextului exista deja creat in PCB
- context switch-ul
  - intre doua threaduri **ale aceluiasi proces:** mai rapid decat context switch-ul a doua procese (nu trebuie schimbat restul contextului din PCB)
  - intre doua threaduri din procese **diferite:** acelasi cost ca si context switch-ul de procese (nu se schimba doar TCB-urile, ci si PCB-urile)

Planificarea kernel threadurilor kernel

- threadurile ruleaza asincron unele fata de celelalte si pot pierde procesorul la fel ca si procesele

  => accesul la date partajate trebuie sincronizat

- planificatorul kernel alege urmatorul thread care trebuie sa ruleze

  => politicile de planificare ale aplicatiei (de ex bazata pe prioritati) trebuie comunicate intr-un fel sau altul kernelului

- pt. multiprocesoare, planificatorul poate aloca mai multe CPU-uri unui singur proces pt. ca threadurile sa ruleze in paralel (*gang scheduling*)
- thread blocat in kernel (de ex prin apel de sistem I/O) + cuanta de timp proces neconsumata integral

  => planificatorul cauta in lista de TCB-uri un thread gata de rulare **din acelasi proces** si ii acorda procesorul

  - consecinta: lightweight context switch, TCB vs PCB

Protectia threadurilor kernel

- observatia generala despre threaduri e valabila si pt threaduri kernel
- un thread poate corupe stiva altui thread, de pilda

  => se distrug datele private ale altui thread (variabilele automatice/locale)

- solutie: implementarea stivelor in spatii de adresa diferite, dar asta mareste costul context switch-ului
  - necesita salvarea si reincarcarea contextelor de executie referitoare la gestiunea memoriei, pe langa contextul uzual din TCB

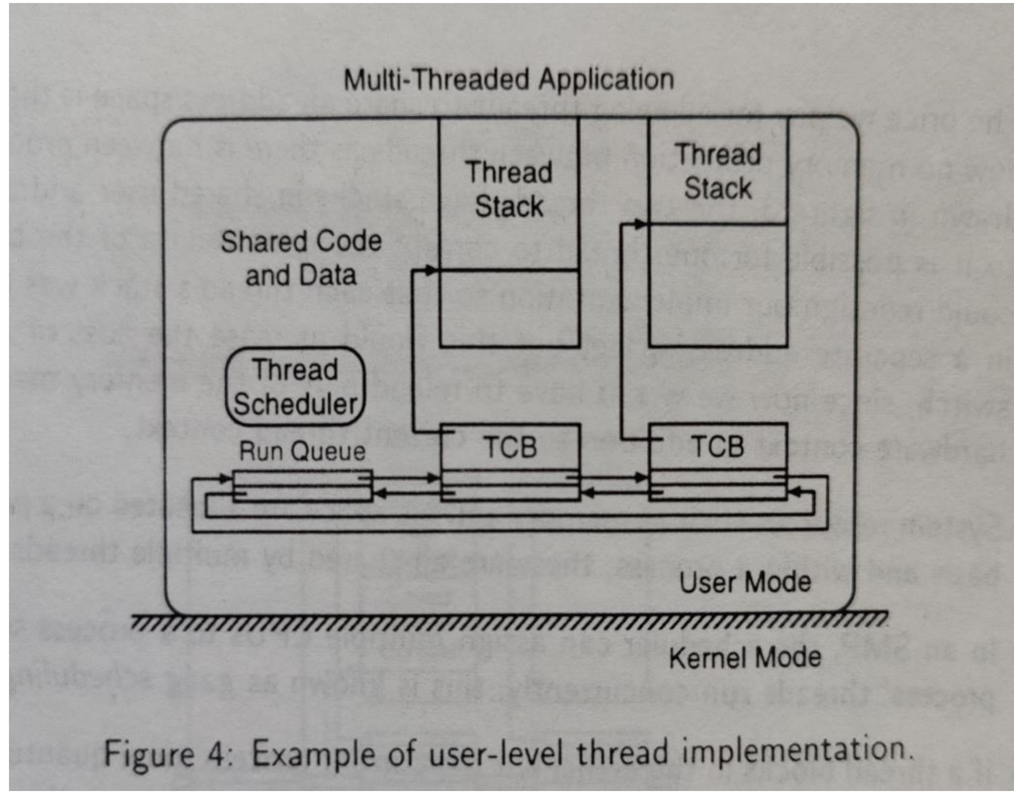
Dezavantajele threadurilor kernel

- desi mai putin costisitoare ca procesele, anumite aspecte le fac nepotrivite pt utilizatori
- (1) *thread_create* este apel sistem => costisitor pt procese care creeaza multe threaduri
- (2) context switching-ul de threaduri necesita intrarea si iesirea in/din kernel mode => overhead aditional context switching-ului obisnuit
- (3) implementarea in kernel e **inflexibila**
  - impune un model de threaduri care nu e potrivit pt orice aplicatie
  - codul planificatorului (scheduler) nu e accesibil (fiind in kernel)

  => greu de adaptat pt cerintele specifice ale unei anumite aplicatii (politica de planificare nu se poate schimba usor)

User-level threads

- daca planificatorul si TCB-urile sunt implementate in spatiul utilizator costurile scad pentru ca nu mai e nevoie de transgresarea granitei kernel/user
- comparatie calitativa, intr-un sistem in care un apel de procedura costa 7 usec, iar un apel sistem (trap) 19 usec



Operatie	Thread user	Thread kernel	Proces Unix
fork	34 usec	948 usec	11300 usec
IPC synch	37 usec	441 usec	1840 usec

Caracteristici threaduri utilizator

- nu apeleaza serviciile kernel pt creare si context switch => aceste operatii sunt f. rapide
  - nu se trece granita user-kernel si nu e nevoie de verificarea parametrilor apelului sistem de catre kernel
- aplicatiile pot furniza propriul planificator (*thread scheduler*) customizat cf unor cerinte specifice
- nu necesita suport explicit din partea kernelului
  - procesul e privit ca un procesor virtual pt. threaduri
- fiind mult mai rapide decat threadurile kernel, de regula threadurile user se implementeaza deasupra threadurilor kernel

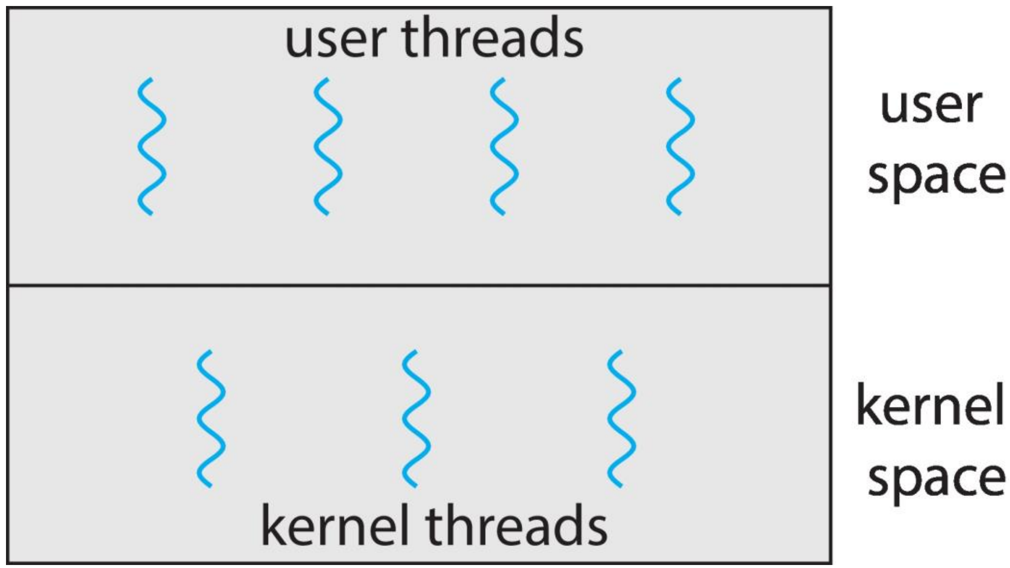
  => planificatorul de threaduri user trateaza threadurile kernel ca pe procesoare virtuale

  - multiplexeaza mai multe threaduri user pe unul sau mai multe threaduri kernel

User Threads and Kernel Threads

- **User threads** - management done by user-level threads library
- Three primary thread libraries:
  - POSIX **Pthreads**
  - Windows threads
  - Java threads
- **Kernel threads** - Supported by the Kernel
- Examples – virtually all general-purpose operating systems, including:
  - Windows
  - Linux
  - Mac OS X
  - iOS
  - Android

User and Kernel Threads

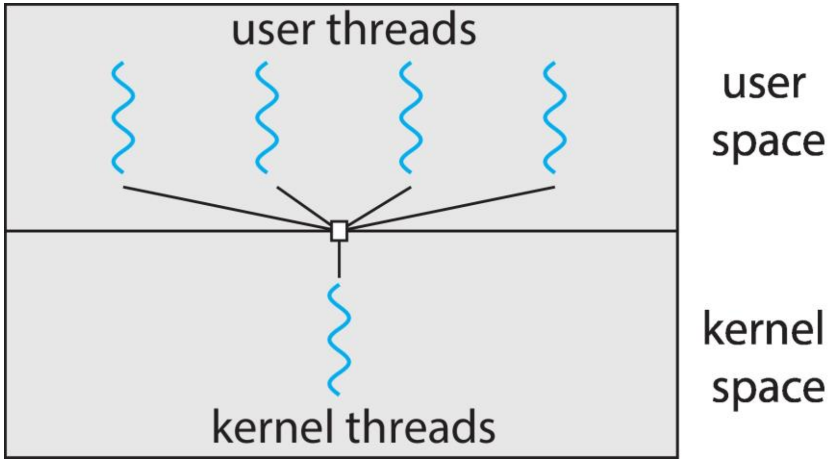


Multithreading Models

- Many-to-One
- One-to-One
- Many-to-Many

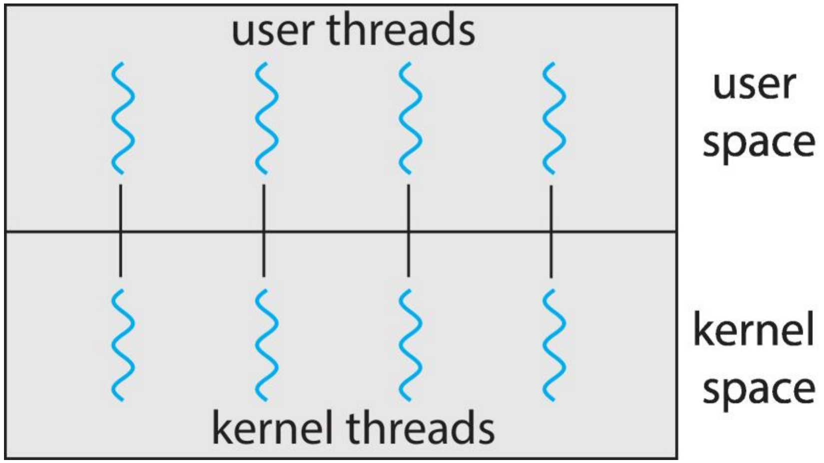
Many-to-One

- Many user-level threads mapped to single kernel thread
- One thread blocking causes all to block
- Multiple threads may not run in parallel on multicore system because only one may be in kernel at a time
- Few systems currently use this model
- Examples:
  - **Solaris Green Threads**
  - **GNU Portable Threads**



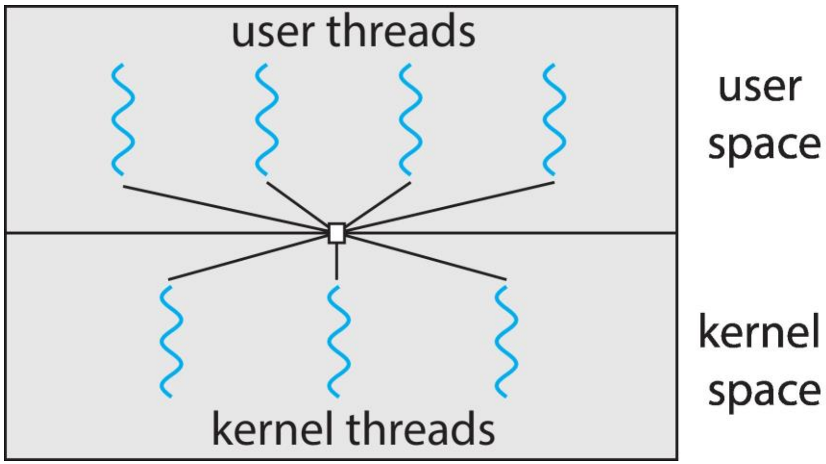
One-to-One

- Each user-level thread maps to kernel thread
- Creating a user-level thread creates a kernel thread
- More concurrency than many-to-one
- Number of threads per process sometimes restricted due to overhead
- Examples
  - Windows
  - Linux



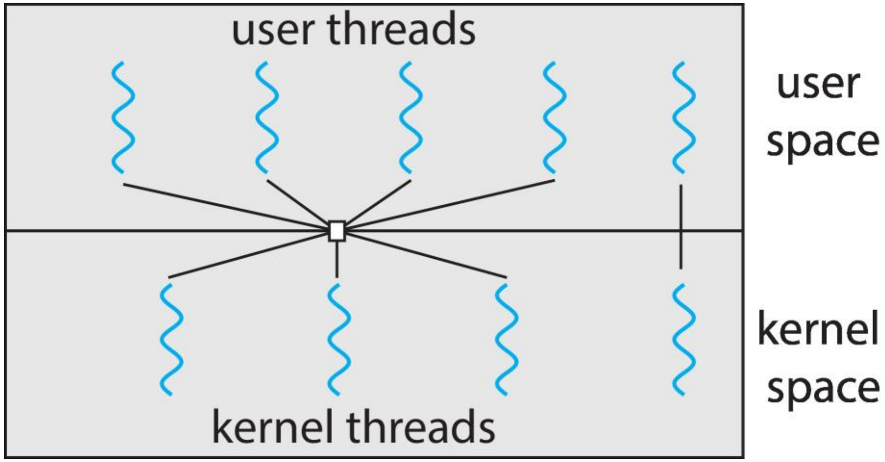
Many-to-Many Model

- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Windows with the *ThreadFiber* package
- Otherwise not very common



Two-level Model

- Similar to M:M, except that it allows a user thread to be **bound** to kernel thread



Probleme comune threadurilor k. si u.

- reentranta: multe apeluri de biblioteca sunt ne-reentrante
  - apelurile reentrante desemnate in manual ca “thread safe”
  - ex: trimiterea unui mesaj in retea in 2 pasi: (a) asamblare mesaj in buffer + (b) transmisie (*syscall*)
  - pierderea procesorului intre (a) si (b) poate conduce la suprascrierea bufferului
  - alte ex: *errno*, *malloc*, apeluri *stdio*
- tratarea semnalelor
  - ex: un thread trateaza un semnal in vreme ce alt thread vrea ca semnalul respectiv sa termine aplicatia (procesul, mai exact)
  - se poate intampla daca se folosesc apeluri de biblioteca si runtime user impreuna
  - cauza: semnalele sunt definite per proces si nu per thread

Dezavantaje threaduri utilizator

- determinate de faptul ca existenta lor e necunoscuta kernelului
- (1) thread user executa apel sistem blocant in kernel
  - planificatorul kernel (agnostic cu privire la threadurile user) considera intreg procesul blocat
  - se aloca CPU altui proces (sau kernel thread), *chiar daca procesul curent mai are si alte threaduri user gata de rulare si nu si-a consumat toata cuanta de timp CPU !*
- (2) thread user comite *page fault* (acces la pagina de memorie inexistenta/nealocata)
  - planificatorul alege alt proces/kernel thread pt executie pe durata transferului paginii de pe disc in memorie

    => aplicatia ruleaza cu mai putine CPU decat e necesar
- (3) kernelul scoate de pe CPU un thread user care detine un (spin)lock ne-eliberat

  => scade dramatic performanta aplicatiilor care folosesc threaduri user in paralel
  - efect exacerbat daca sunt alese sa ruleze alte threaduri care vor sa obtina acelasi spinlock
- *problema de fond*: lipsa de coordonare scheduler kernel <–> pachet de threaduri user

Scheduler activations

- metoda de coordonare kernel <–> pachet de threaduri utilizator
- model: aplicatia ruleaza pe un multiprocesor virtual
  - apeluri sistem pt a suplimenta/diminua nr de procesoare virtuale alocate de kernel (“adauga CPU”, “acest CPU este idle”)
  - kernelul decide daca onoreaza cererea sau nu
- scheduler activation e aproximarea unui kernel thread
  - are stive kernel si user
  - ofera context de executie pt un thread user
  - in plus, ofera conceptul de *upcall* pt evenimente de mai multe tipuri
    - fiecare upcall creeaza o noua activare (optimizare: folosirea vechilor activari)

Tipuri de evenimente upcall

- *adauga procesor* – consecinta: executia unui thread user
- *procesor preemptat* – adauga threadul user care se executa in activarea care a pierdut CPU in coada de threaduri gata de rulare
- *activare blocata* – activarea s-a blocat (eg, apel sistem blocant) si nu mai utilizeaza procesorul
- *activare deblocata* – pune in lista gata de rulare threadul care se executa in contextul activarii blocate

Functionarea scheduler activations

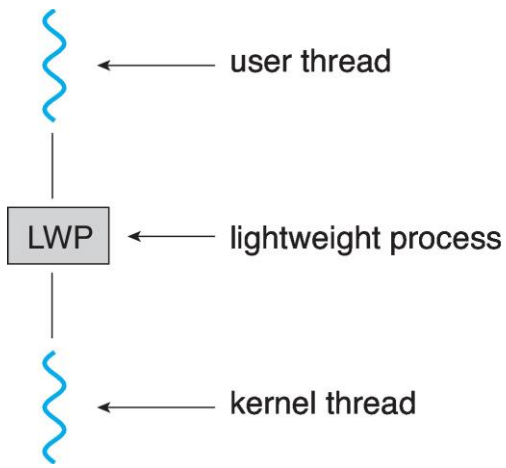
- la pornirea procesului
  - kernelul aloca o activare + notifica aplicatia (*upcall* “adauga CPU”) dupa ce i-a asigurat un CPU
  - sistemul de gestiune al threadurilor user primeste notificarea
    - foloseste noua activare drept context de executie pt initializarea sa si a threadului *main*
    - ulterior se pot crea noi threaduri si cere noi procesoare
- la cerere de suplimentare a concurentei (adaugare CPU sau pt. thread blocat)
  - kernelul salveaza starea threadului user in activarea curenta
  - aloca o noua activare si cheama aplicatia in contextul noii activari

Functionarea scheduler activations (cont)

- la blocarea unui thread user in kernel
  - nu se continua in kernel la deblocare
  - se creeaza o noua activare
  - se notifica aplicatia
  - schedulerul user copiaza starea threadului blocat din vechea activare si notifica kernelul ca vechea activare poate fi refolosita
- cand un thread user care detine un spinlock pierde procesorul
  - kernelul genereaza un *upcall*
  - sistemul de gestiune a threadurilor verifica daca threadul e in sectiune critica
  - in caz afirmativ, threadul este continuat temporar printr-un context switch in user space
  - la iesirea din sectiunea critica, se da controlul inapoi *upcall-ului* tot prin context switch in user space

Scheduler Activations

- Both M:M and Two-level models require communication to maintain the appropriate number of kernel threads allocated to the application
- Typically use an intermediate data structure between user and kernel threads – **lightweight process** (**LWP**)
  - Appears to be a virtual processor on which process can schedule user thread to run
  - Each LWP attached to kernel thread
  - How many LWPs to create?
- Scheduler activations provide **upcalls** - a communication mechanism from the kernel to the **upcall handler** in the thread library
- This communication allows an application to maintain the correct number kernel threads



Thread Libraries

- **Thread library** provides programmer with API for creating and managing threads
- Two primary ways of implementing
  - Library entirely in user space
  - Kernel-level library supported by the OS

Exemple de pachete de threaduri user

- Solaris si POSIX
- threaduri Solaris
  - exista si la nivel de kernel (s.n. LWPs, sunt multiplexate pe CPU existente) si la nivel de utilizator
  - pachetul de threaduri (*libthread*) planifica threadurile user pe o colectie de LWPs
  - uzual, threadurile user sunt create nelegate (*unbound*), adica se pot muta intre LWP-uri si asa se si folosesc in general
  - daca un thread user e asignat unui LWP se spune ca e legat (*bound*)
  - controlul kernelului asupra threadurilor user se exercita prin bound threads cu ajutorul setarii prioritatii LWP-ului asociat (de ex, thread user pt stream audio cu prioritate mare in aplicatie multimedia)

Thread-uri Solaris (cont.)

- ….
  - biblioteca de threaduri asigura ca exista suficiente LWP-uri active pt ca procesul sa poata continua
  - daca un proces determina toate LWP-urile sale sa se blocheze, biblioteca va crea LWP-uri aditionale pt. ca threadurile neblocate sa poata rula
  - daca LWP-urile sunt idle prea mult timp, biblioteca le da inapoi kernelului pt a fi folosite de alte aplicatii

Thread-uri Solaris (cont.)

- exista apel de setare explicita a nr de LWP-uri la un nou nivel

  thr_setconcurrency(int new_level)

- mecanisme IPC: mutex si variabile conditie
- implementeaza TLS (Thread-Local Storage), capacitate privata de stocare a variabilelor in afara stivei
  - variabile locale nepartajate, “statice” (globale pt threadul respectiv)
  - declarate cu #pragma, de ex #pragma unshared errno;

Solaris threads

```
int thr_create(void *stack_base, size_t stack_size,
               void *(*function)(void *), void *arg, long flags)
void thr_exit(void *status)
int thr_join(thread_t wait_for, thread_t *joiner_id, void *status)
int mutex_init(mutex_t *mutex, int type, void *arg)
int mutex_lock(mutex_t *mutex)
int mtex_unlock(mutex_t *mutex);
int cond_init(cond_t *cond, int type, int arg)
int cond_wait(cond_t *cond, mutex_t *mutex)
int cond_signal(cond_t *cond)
int cond_broadcast(cond_t *cond)
```

Figure 5: Some commonly used UI thread operations.

Tratarea semnalelor

- fiecare thread are propria masca de semnale
- toate threadurile unui proces partajeaza handlerele de semnal
- cand SIG_IGN/SIG_DFL sunt setate, se aplica tuturor threadurilor procesului
- semnalele de tip *trap* (sincrone cu executia threadului, ex SIGFPE, SIGSEGV) executate exclusiv de threadul care le-a generat

  => m. m. threaduri pot genera si trata acelasi tip de semnal simultan
- semnalele de tip *intrerupere* (generate asincron cu executia threadului, ex SIGIO, SIGINT)
  - pot fi tratate de orice thread care nu mascheaza/blocheaza semnalul respectiv
  - dc. sunt mai multe astfel de threaduri, se alege unul dintre ele pt tratarea semnalului
- daca toate threadurile au mascat un anume semnal, se asteapta dupa primul thread care deblocheaza tratarea semnalului respectiv

Tratarea semnalelor (cont.)

- *thread_kill*: trimite un semnal catre un thread din acelasi proces
  - semnalul e de tip trap, tratat doar de threadul caruia ii este adresat
  - nu se pot trimite semnale unui *anume* thread dintr-un *alt* proces
- threadurile Solaris implementeaza un nou semnal, SIGWAITING (SIG_IGN implicit)
  - trimis procesului cand toate LWP-urile sunt blocate indefinit in asteptarea unui eveniment extern
  - poate fi folosit pt crearea de noi LWP-uri pt a evita blocarea intregului proces
  - idee similara cu scheduler activations, dar coarse-grain (nu merge pt short-term blocking, doar pt evenimente de tip page faults, filesystem I/O)

Thread-uri POSIX

- *pthreads*
- standard IEEE, Portable Operating System Interface (e o specificatie, nu o implementare)
- asemanatoare threadurilor Solaris

Pthreads

- May be provided either as user-level or kernel-level
- A POSIX standard (IEEE 1003.1c) API for thread creation and synchronization
- ***Specification***, not ***implementation***
- API specifies behavior of the thread library, implementation is up to development of the library
- Common in UNIX operating systems (Linux & Mac OS X)

POSIX threads

```
int pthread_create(pthread_t *thread_pointer,
                   pthread_attr_t *attributes,
                    void *(*function)(void *), void *arg)
void pthread_exit(void *status)
int pthread_join(pthread_t thread, void **status)
void pthread_mutex_init(pthread_mutex_t *mutex,
                        pthread_mutexattr_t *attr)
int pthread_mutex_lock(pthread_mutex_t *mutex)
int pthread_mutex_unlock(pthread_mutex_t *mutex)
int pthread_cond_init(pthread_cond_t *cond, pthread_condattr_t *attr)
int pthread_cond_wait(pthread_cond_t *cond, pthread_mutex_t *mutex)
int pthread_cond_signal(pthread_cond_t *cond)
int pthread_cond_broadcast(pthread_cond_t *cond)
```

Figure 6: Some commonly used Pthreads operations.

Pthreads Example

```
#include <pthread.h>
#include <stdio.h>

#include <stdlib.h>

int sum; /* this data is shared by the thread(s) */
void *runner(void *param); /* threads call this function */

int main(int argc, char *argv[])
{
   pthread_t tid; /* the thread identifier */
   pthread_attr_t attr; /* set of thread attributes */

   /* set the default attributes of the thread */
   pthread_attr_init(&attr);
   /* create the thread */
   pthread_create(&tid, &attr, runner, argv[1]);
   /* wait for the thread to exit */
   pthread_join(tid,NULL);

   printf("sum = %d\n",sum);
}
```

Pthreads Example (Cont.)

```
/* The thread will execute in this function */
void *runner(void *param)
{
   int i, upper = atoi(param);
   sum = 0;

   for (i = 1; i <= upper; i++)
      sum += i;

   pthread_exit(0);
}
```

Pthreads Code for Joining 10 Threads

```
#define NUM_THREADS 10

/* an array of threads to be joined upon */
pthread_t workers[NUM_THREADS];

for (int i = 0; i < NUM_THREADS; i++)
   pthread_join(workers[i], NULL);
```

Implicit Threading

- Growing in popularity as numbers of threads increase, program correctness more difficult with explicit threads
- Creation and management of threads done by compilers and run-time libraries rather than programmers
- Five methods explored
  - Thread Pools
  - Fork-Join
  - OpenMP
  - Grand Central Dispatch
  - Intel Threading Building Blocks

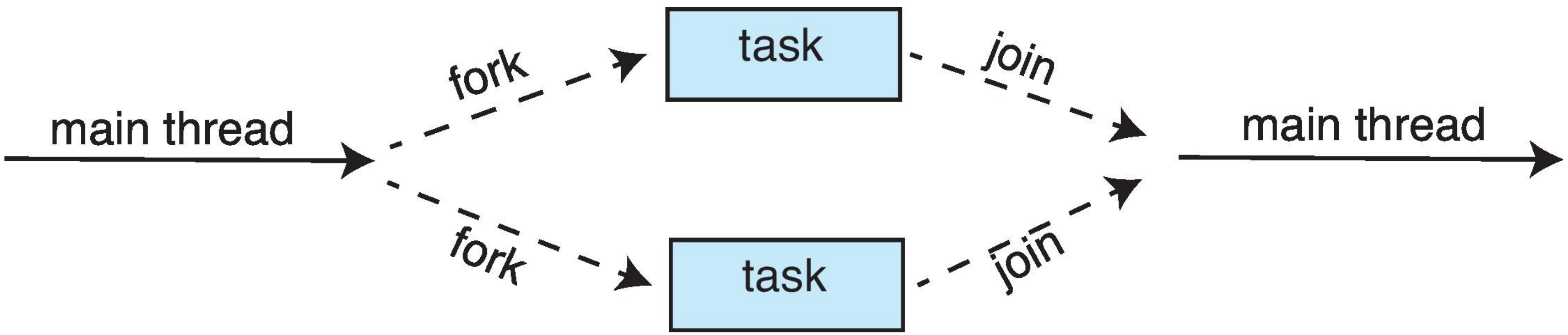
Thread Pools

- Create a number of threads in a pool where they await work
- Advantages:
  - Usually slightly faster to service a request with an existing thread than create a new thread
  - Allows the number of threads in the application(s) to be bound to the size of the pool
  - Separating task to be performed from mechanics of creating task allows different strategies for running task
    - i.e,Tasks could be scheduled to run periodically
- Windows API supports thread pools:

```
DWORD WINAPI PoolFunction(AVOID Param) {
    /*
    * this function runs as a separate thread.
    */
}
```

Fork-Join Parallelism

- Multiple threads (tasks) are **forked**, and then **joined**.

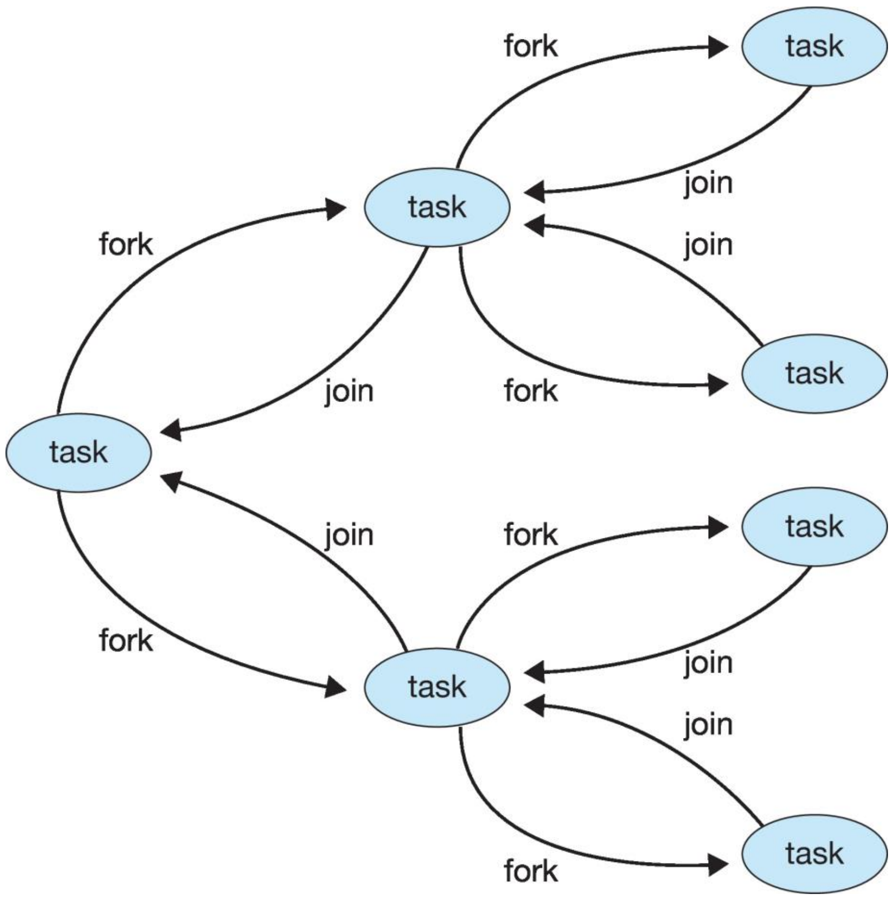


Fork-Join Parallelism

- General algorithm for fork-join strategy:

```
Task(problem)
   if problem is small enough
      solve the problem directly
   else
      subtask1 = fork(new Task(subset of problem)
      subtask2 = fork(new Task(subset of problem)

      result1 = join(subtask1)
      result2 = join(subtask2)

      return combined results
```

Fork-Join Parallelism



Threading Issues

- Semantics of **fork()** and **exec()** system calls
- Signal handling
  - Synchronous and asynchronous
- Thread cancellation of target thread
  - Asynchronous or deferred
- Thread-local storage
- Scheduler Activations

Semantics of fork() and exec()

- Does **`fork()`** duplicate only the calling thread or all threads?
  - Some UNIXes have two versions of fork
- **`exec()`** usually works as normal – replace the running process including all threads

Signal Handling

- **Signals** are used in UNIX systems to notify a process that a particular event has occurred.
- A **signal handler** is used to process signals
  1. Signal is generated by particular event
  2. Signal is delivered to a process
  3. Signal is handled by one of two signal handlers:
     1. default
     2. user-defined
- Every signal has **default handler** that kernel runs when handling signal
  - **User-defined signal handler** can override default
  - For single-threaded, signal delivered to process

Signal Handling (Cont.)

- Where should a signal be delivered for multi-threaded?
  - Deliver the signal to the thread to which the signal applies
  - Deliver the signal to every thread in the process
  - Deliver the signal to certain threads in the process
  - Assign a specific thread to receive all signals for the process

Thread Cancellation

- Terminating a thread before it has finished
- Thread to be canceled is **target thread**
- Two general approaches:
  - **Asynchronous cancellation** terminates the target thread immediately
  - **Deferred cancellation** allows the target thread to periodically check if it should be cancelled
- Pthread code to create and cancel a thread:

```
pthread_t tid;

/* create the thread */
pthread_create(&tid, 0, worker, NULL);

 . . .

/* cancel the thread */
pthread_cancel(tid);

/* wait for the thread to terminate */
pthread_join(tid,NULL);
```

Thread Cancellation (Cont.)

- Invoking thread cancellation requests cancellation, but actual cancellation depends on thread state

Mode	State	Type
Off	Disabled	–
Deferred	Enabled	Deferred
Asynchronous	Enabled	Asynchronous

- If thread has cancellation disabled, cancellation remains pending until thread enables it
- Default type is deferred
  - Cancellation only occurs when thread reaches **cancellation point**
    - i.e., `pthread_testcancel()`
    - Then **cleanup handler** is invoked
- On Linux systems, thread cancellation is handled through signals

Thread-Local Storage

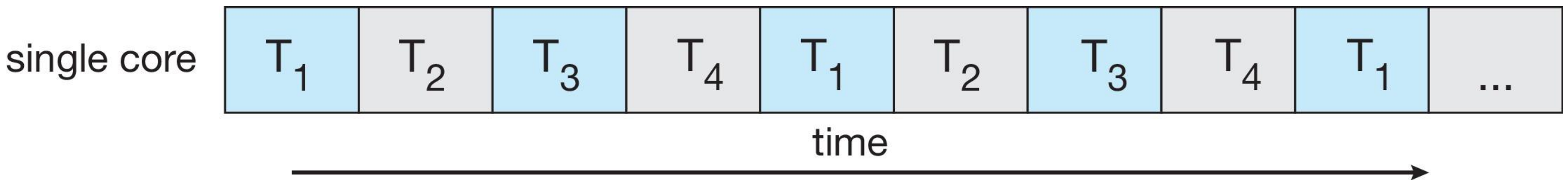
- **Thread-local storage** (**TLS**) allows each thread to have its own copy of data
- Useful when you do not have control over the thread creation process (i.e., when using a thread pool)
- Different from local variables
  - Local variables visible only during single function invocation
  - TLS visible across function invocations
- Similar to `static` data
  - TLS is unique to each thread

Multicore Programming

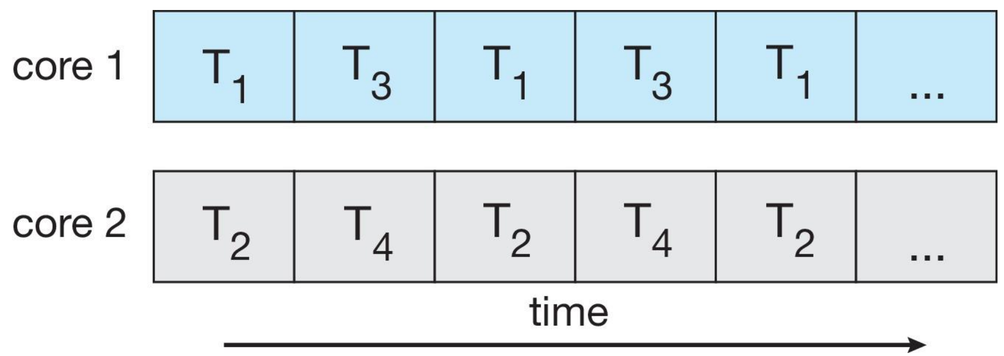
- **Multicore** or **multiprocessor** systems puts pressure on programmers, challenges include:
  - **Dividing activities**
  - **Balance**
  - **Data splitting**
  - **Data dependency**
  - **Testing and debugging**
- ***Parallelism*** implies a system can perform more than one task simultaneously
- ***Concurrency*** supports more than one task making progress
  - Single processor / core, scheduler providing concurrency

Concurrency vs. Parallelism

- Concurrent execution on single-core system:



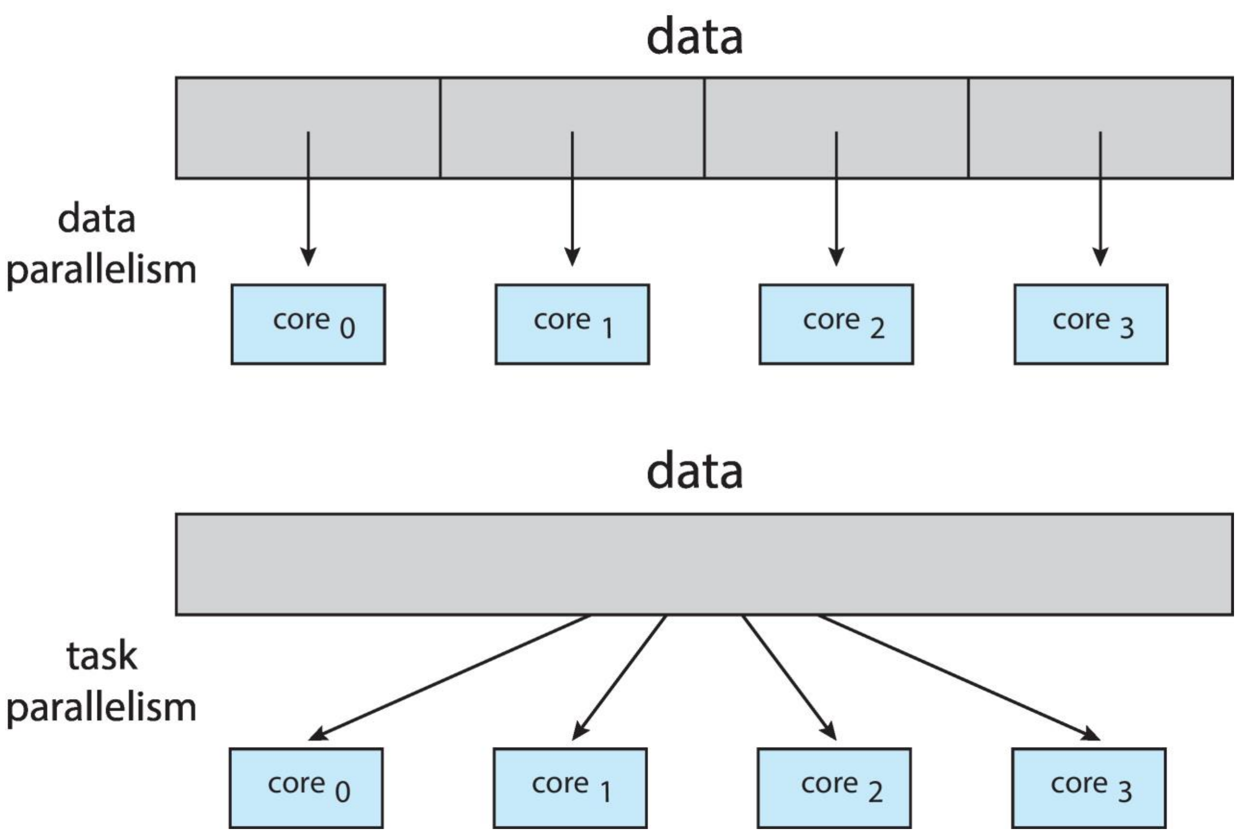
- Parallelism on a multi-core system:



Multicore Programming

- Types of parallelism
  - **Data parallelism** – distributes subsets of the same data across multiple cores, same operation on each
  - **Task parallelism** – distributing threads across cores, each thread performing unique operation

Data and Task Parallelism



Amdahl's Law

- Identifies performance gains from adding additional cores to an application that has both serial and parallel components
- *S* is serial portion
- *N* processing cores

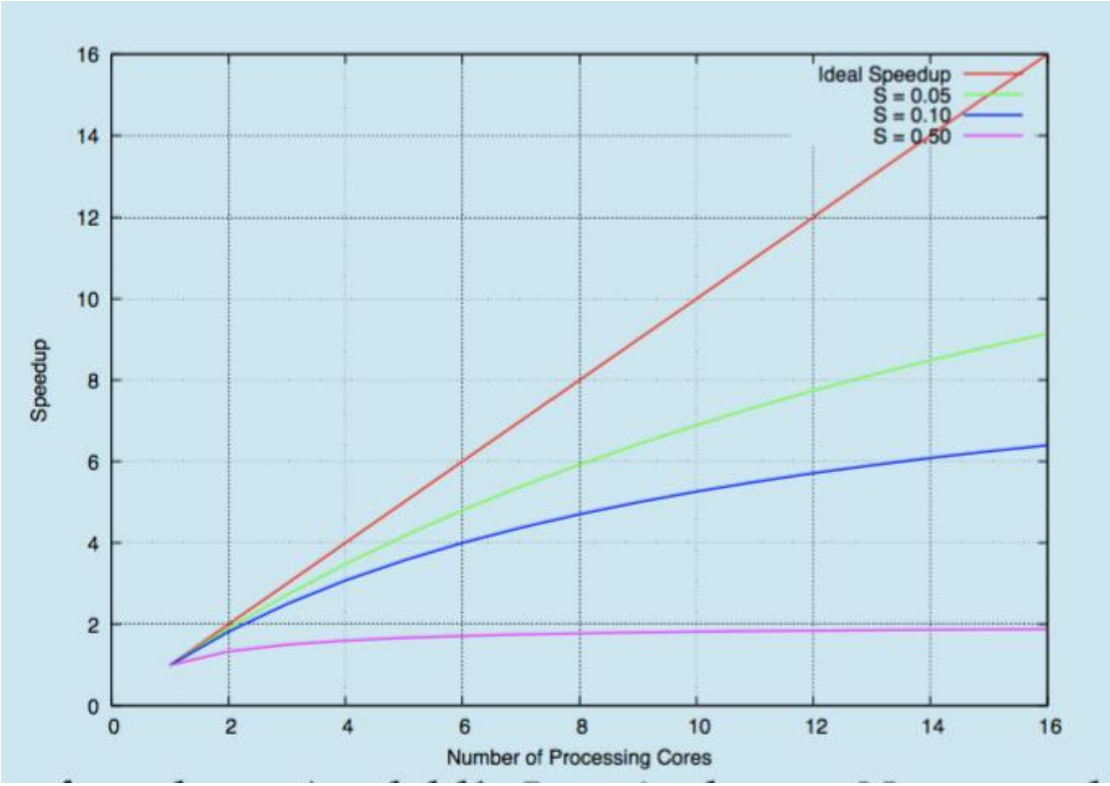
$$speedup \leq \frac{1}{S + \frac{(1-S)}{N}}$$

- That is, if application is 75% parallel / 25% serial, moving from 1 to 2 cores results in speedup of 1.6 times
- As *N* approaches infinity, speedup approaches 1 / *S*

Serial portion of an application has disproportionate effect on performance gained by adding additional cores

- But does the law take into account contemporary multicore systems?

Amdahl's Law


Ideal Speedup
S = 0.05
S = 0.10
S = 0.50
Speedup
Number of Processing Cores
0
2
4
6
8
10
12
14
16

Windows Multithreaded C Program

```
#include <windows.h>
#include <stdio.h>
DWORD Sum; /* data is shared by the thread(s) */

/* The thread will execute in this function */
DWORD WINAPI Summation(LPVOID Param)
{
   DWORD Upper = *(DWORD*)Param;
   for (DWORD i = 1; i <= Upper; i++)
      Sum += i;
   return 0;
}
```

Windows Multithreaded C Program (Cont.)

```
int main(int argc, char *argv[])
{
  DWORD ThreadId;
  HANDLE ThreadHandle;
  int Param;

  Param = atoi(argv[1]);
  /* create the thread */
  ThreadHandle = CreateThread(
    NULL, /* default security attributes */
    0, /* default stack size */
    Summation, /* thread function */
    &Param, /* parameter to thread function */
    0, /* default creation flags */
    &ThreadId); /* returns the thread identifier */

   /* now wait for the thread to finish */
  WaitForSingleObject(ThreadHandle,INFINITE);

  /* close the thread handle */
  CloseHandle(ThreadHandle);

  printf("sum = %d\n",Sum);
}
```

Java Threads

- Java threads are managed by the JVM
- Typically implemented using the threads model provided by underlying OS
- Java threads may be created by:
  - Extending Thread class
  - Implementing the Runnable interface

```
public interface Runnable
{
    public abstract void run();
}
```

  - Standard practice is to implement Runnable interface

Java Threads

Implementing Runnable interface:

```
class Task implements Runnable
{
   public void run() {
      System.out.println("I am a thread.");
   }
}
```

Creating a thread:

```
Thread worker = new Thread(new Task());
worker.start();
```

Waiting on a thread:

```
try {
   worker.join();
}
catch (InterruptedException ie) { }
```

Java Executor Framework

- Rather than explicitly creating threads, Java also allows thread creation around the Executor interface:

```
public interface Executor
{
   void execute(Runnable command);
}
```

- The Executor is used as follows:

```
Executor service = new Executor;
service.execute(new Task());
```

Java Executor Framework

```
import java.util.concurrent.*;

class Summation implements Callable<Integer>
{
   private int upper;
   public Summation(int upper) {
      this.upper = upper;
   }

   /* The thread will execute in this method */
   public Integer call() {
      int sum = 0;
      for (int i = 1; i <= upper; i++)
         sum += i;

      return new Integer(sum);
   }
}
```

Java Executor Framework (Cont.)

```
public class Driver
{
 public static void main(String[] args) {
    int upper = Integer.parseInt(args[0]);

    ExecutorService pool = Executors.newSingleThreadExecutor();
    Future<Integer> result = pool.submit(new Summation(upper));

    try {
       System.out.println("sum = " + result.get());
    } catch (InterruptedException | ExecutionException ie) { }
  }
}
```

Java Thread Pools

- Three factory methods for creating thread pools in Executors class:
  - `static ExecutorService newSingleThreadExecutor()`
  - `static ExecutorService newFixedThreadPool(int size)`
  - `static ExecutorService newCachedThreadPool()`

Java Thread Pools (Cont.)

```
import java.util.concurrent.*;

public class ThreadPoolExample
{
public static void main(String[] args) {
   int numTasks = Integer.parseInt(args[0].trim());

   /* Create the thread pool */
   ExecutorService pool = Executors.newCachedThreadPool();

   /* Run each task using a thread in the pool */
   for (int i = 0; i < numTasks; i++)
      pool.execute(new Task());

   /* Shut down the pool once all threads have completed */
   pool.shutdown();
}
```

Fork-Join Parallelism in Java

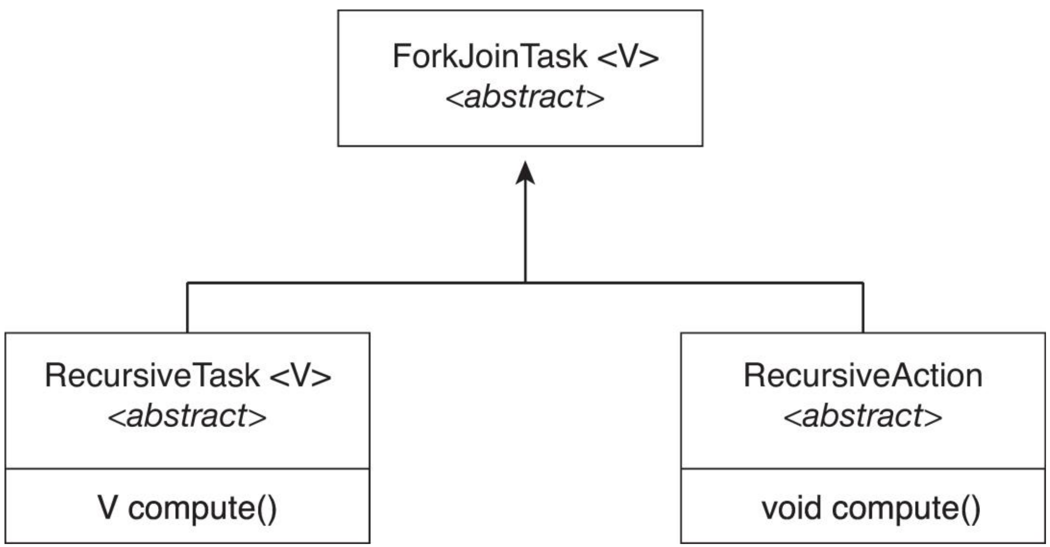
```
ForkJoinPool pool = new ForkJoinPool();
// array contains the integers to be summed
int[] array = new int[SIZE];

SumTask task = new SumTask(0, SIZE - 1, array);
int sum = pool.invoke(task);
```

Fork-Join Parallelism in Java

```
import java.util.concurrent.*;

public class SumTask extends RecursiveTask<Integer>
{
   static final int THRESHOLD = 1000;

   private int begin;
   private int end;
   private int[] array;

   public SumTask(int begin, int end, int[] array) {
      this.begin = begin;
      this.end = end;
      this.array = array;
   }

   protected Integer compute() {
      if (end - begin < THRESHOLD) {
         int sum = 0;
         for (int i = begin; i <= end; i++)
            sum += array[i];

         return sum;
      }
      else {
         int mid = (begin + end) / 2;

         SumTask leftTask = new SumTask(begin, mid, array);
         SumTask rightTask = new SumTask(mid + 1, end, array);

         leftTask.fork();
         rightTask.fork();

         return rightTask.join() + leftTask.join();
      }
   }
}
```

Fork-Join Parallelism in Java

- The **`ForkJoinTask`** is an abstract base class
- **`RecursiveTask`** and **`RecursiveAction`** classes extend **`ForkJoinTask`**
- **`RecursiveTask`** returns a result (via the return value from the **`compute()`** method)
- **`RecursiveAction`** does not return a result



OpenMP

- Set of compiler directives and an API for C, C++, FORTRAN
- Provides support for parallel programming in shared-memory environments
- Identifies **parallel regions** – blocks of code that can run in parallel

`#pragma omp parallel`

Create as many threads as there are cores

```
#include <omp.h>
#include <stdio.h>

int main(int argc, char *argv[])
{
  /* sequential code */

  #pragma omp parallel
  {
    printf("I am a parallel region.");
  }

  /* sequential code */

  return 0;
}
```

- Run the for loop in parallel

```
#pragma omp parallel for
for (i = 0; i < N; i++) {
   c[i] = a[i] + b[i];
}
```

Grand Central Dispatch

- Apple technology for macOS and iOS operating systems
- Extensions to C, C++ and Objective-C languages, API, and run-time library
- Allows identification of parallel sections
- Manages most of the details of threading
- Block is in “^{ }” :

```
^{ printf("I am a block"); }
```

- Blocks placed in dispatch queue
  - Assigned to available thread in thread pool when removed from queue

Grand Central Dispatch

- Two types of dispatch queues:
  - **serial** – blocks removed in FIFO order, queue is per process, called **main queue**
    - Programmers can create additional serial queues within program
  - **concurrent** – removed in FIFO order but several may be removed at a time
    - Four system wide queues divided by quality of service:
      - `QOS_CLASS_USER_INTERACTIVE`
      - `QOS_CLASS_USER_INITIATED`
      - `QOS_CLASS_USER_UTILITY`
      - `QOS_CLASS_USER_BACKGROUND`

Grand Central Dispatch

- For the Swift language a task is defined as a closure – similar to a block, minus the caret
- Closures are submitted to the queue using the `dispatch_async()` function:

```
let queue = dispatch_get_global_queue
   (QOS_CLASS_USER_INITIATED, 0)

dispatch_async(queue,{ print("I am a closure.") })
```

Intel Threading Building Blocks (TBB)

- Template library for designing parallel C++ programs
- A serial version of a simple for loop

```
for (int i = 0; i < n; i++) {
   apply(v[i]);
}
```

- The same for loop written using TBB with **`parallel_for`** statement:

```
parallel_for (size_t(0), n, [=](size_t i) {apply(v[i]);});
```

Thread Cancellation in Java

- Deferred cancellation uses the **`interrupt()`** method, which sets the interrupted status of a thread.

```
Thread worker;

 . . .

/* set the interruption status of the thread */
worker.interrupt()
```

- A thread can then check to see if it has been interrupted:

```
while (!Thread.currentThread().isInterrupted()) {
    . . .
}
```

Operating System Examples

- Windows Threads
- Linux Threads

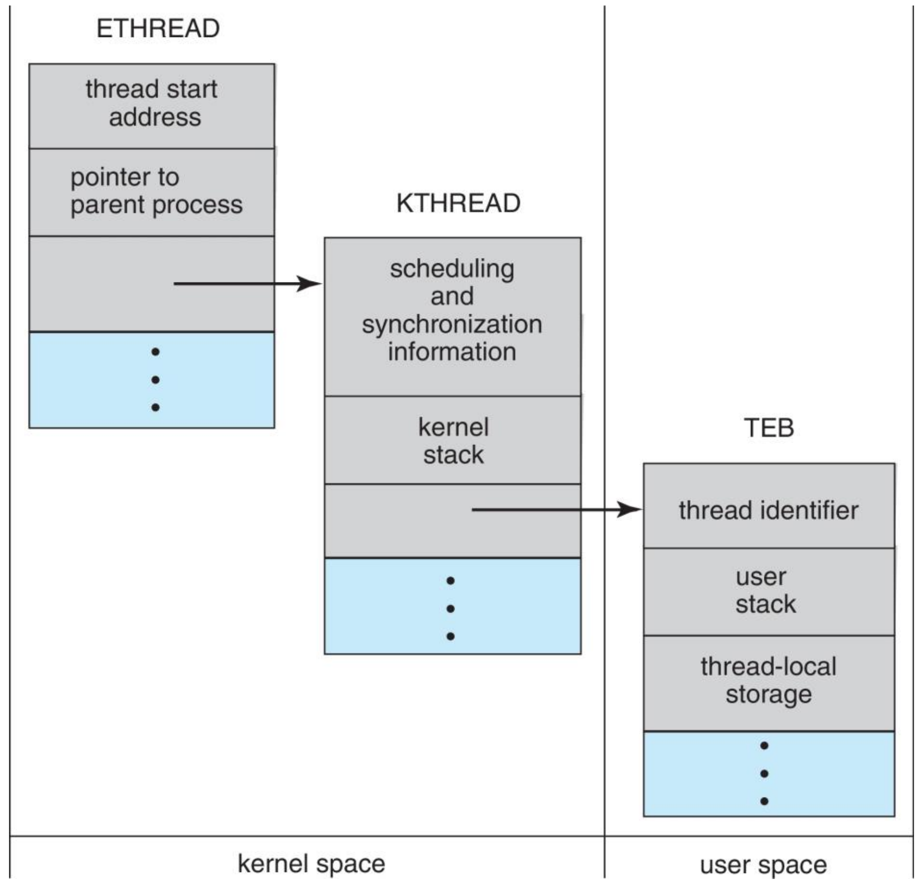
Windows Threads

- Windows API – primary API for Windows applications
- Implements the one-to-one mapping, kernel-level
- Each thread contains
  - A thread id
  - Register set representing state of processor
  - Separate user and kernel stacks for when thread runs in user mode or kernel mode
  - Private data storage area used by run-time libraries and dynamic link libraries (DLLs)
- The register set, stacks, and private storage area are known as the **context** of the thread

Windows Threads (Cont.)

- The primary data structures of a thread include:
  - ETHREAD (executive thread block) – includes pointer to process to which thread belongs and to KTHREAD, in kernel space
  - KTHREAD (kernel thread block) – scheduling and synchronization info, kernel-mode stack, pointer to TEB, in kernel space
  - TEB (thread environment block) – thread id, user-mode stack, thread-local storage, in user space

Windows Threads Data Structures



Linux Threads

- Linux refers to them as ***tasks*** rather than ***threads***
- Thread creation is done through **`clone()`** system call
- **`clone()`** allows a child task to share the address space of the parent task (process)
  - Flags control behavior

flag	meaning
`CLONE_FS`	File-system information is shared.
`CLONE_VM`	The same memory space is shared.
`CLONE_SIGHAND`	Signal handlers are shared.
`CLONE_FILES`	The set of open files is shared.

- **`struct task_struct`** points to process data structures (shared or unique)

End of Chapter 4